

Thin-film lithium niobate quantum photonics: review and perspectives

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Abstract. Photonics has proven to be a very attractive platform for quantum technologies, offering key features such as high-fidelity qubits and room-temperature signal processing. Advancements in integrated photonics are expected to further enhance these capabilities as the technology evolves from few-photon architectures to systems capable of generating and processing tens, possibly hundreds, of photons, marking critical progress toward scalable quantum information processing. Although each integrated platform has its own unique advantages and limitations, thin-film lithium niobate (TFLN) photonics has recently emerged as a strong contender thanks to its low-loss characteristics, large electro-optic and nonlinear coefficients, broad transparency window, and ultra-fast modulation capabilities. In this review, we examine the latest developments in TFLN quantum photonics and identify promising directions and challenges for future research in this field.

Keywords: quantum optics; integrated photonics; thin-film lithium niobate; lithium niobate on insulator; low loss; high speed; nonlinearity; quantum frequency conversion; squeezing state; photon pairs; multiplexing; quantum metrology; quantum network; quantum computing.

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1 Introduction

Quantum photonics is set to play a crucial role in advancing information technologies, particularly in the fields of quantum computing, communication, and metrology. Quantum photonics systems may outperform classical supercomputers for certain tasks¹ and may also be positioned to enable long-range quantum networks,² for interconnecting multiple quantum devices,³ for implementing error correction,⁴ for building quantum circuits for computations and simulations,^{5,6} and for generating essential resources for universal quantum computation and communication.⁷ It also plays a crucial role in precision metrology, facilitating multiparameter estimation⁸ and enhancing sensitivity in applications such as gravitational wave detection.⁹

Scaling these quantum systems up is essential for enhancing computational power, improving measurement precision, and extending communication distances. However, the inherent complexity and sensitivity of these systems present significant

challenges in achieving scalability. Integrated quantum photonics offers a promising path forward, providing compact and scalable platforms that reduce size, weight, and power consumption compared with traditional free-space optics systems. Integrated quantum photonics platforms also offer phase-stable optical paths, which help minimize environmental disruptions. Even though manufacturing imperfections for a quantum chip are unavoidable, they manifest as static errors rather than stochastic noise, meaning that they do not cause photonic states to evolve randomly if modeled precisely. These static errors can be effectively characterized using classical light,¹⁰ enabling high precision in system performance.

Photonics is uniquely suited for quantum networks, where the goal is the reliable transmission of quantum states between distant parties. The best-known example is quantum key distribution (QKD), which enables the secure sharing of cryptographic keys.¹¹ To achieve higher secret key rates, QKD systems must incorporate low-loss modulators operating at high clock rates, efficient fiber-to-chip couplers, and high-efficiency detectors. Moreover, future quantum networks aiming to transmit quantum information globally, i.e., a global quantum internet,¹²

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require the distribution of photons over long distances (more than 100 km) among multiple remote nodes, a task hindered by photon loss that scales exponentially with the channel length. Quantum repeaters¹³ equipped with matter quantum memories are therefore crucial as quantum information cannot be amplified or cloned due to the no-cloning theorem.¹⁴ On the other hand, all-optical quantum repeaters,¹⁵ which remove the necessity of matter quantum memory, require large entangled resource states,¹⁶ i.e., quantum systems that are nonclassically correlated across many particles. These resource states can be built using fast active feedforwarding techniques in linear optics, i.e., multiplexing, which require high-speed and low-loss electro-optic switches.

Quantum metrology,^{17,18} with its ability to surpass classical measurement limits, also stands to benefit significantly from integrated photonics as it requires stability against environmental factors, which can interfere with measurement accuracy. Key properties of the material such as thermal stability and high nonlinearity are essential for enhancing measurement precision. Furthermore, many physical processes involve multiple parameters to estimate, making large-scale integration crucial for advancing quantum metrology. In practical applications, integrated quantum metrology could lead to portable, high-precision instruments for fields such as environmental monitoring

and biomedical diagnostics¹⁹—capabilities limited to specialized laboratories so far.

The ultimate goal for quantum information processing is to achieve universal fault-tolerant photonic quantum computing, a challenge of immense complexity. As progress continues, focusing on intermediate milestones is considered fruitful. Currently, we are in the era of noisy intermediate-scale quantum (NISQ) technologies,²⁰ where the goal is to surpass classical supercomputers in areas such as quantum simulations,²¹ optimization problems,²² and particular tasks that challenge classical algorithms,²³ utilizing experimental capabilities that are currently within reach. However, expanding quantum computing power to tackle more complex NISQ problems requires integrating thousands of optical components on-chip, which introduces potential losses and scaling complexity. In addition, the advancement of large-scale quantum information technologies relies on the periodic, near deterministic, or at least highly predictable generation of single photons as these technologies need many single-photon inputs in different modes simultaneously. A promising approach involves multiplexing to convert heralded photons into on-demand photons, by detecting the idler photon to herald the presence of a signal photon and relocating the signal photons to a predetermined target output mode through fast active feedforwarding.^{24–26} Achieving this requires a low-loss

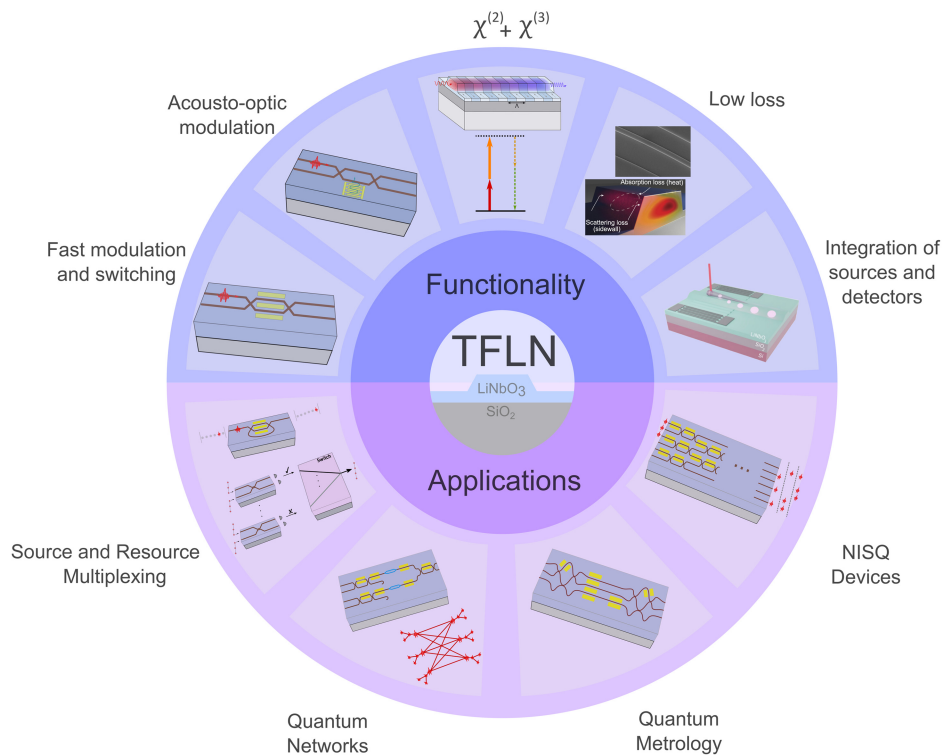


Fig. 1 Thin-film lithium niobate functionality versus potential quantum photonics applications: TFLN gives a big various functionality based on the fundamental material properties and technological development, i.e., acousto-optic modulators and transducers, electro-optic modulation and switches, second- and third-order nonlinearity (waveguide Figure adapted with permission from Ref. 27 © 2023 Optica Publishing Group), low loss (Figure adapted with permission from Ref. 28 © 2022 The Author(s)), and sources and detectors integration (Figure adapted with permission from Ref. 29 © 2023 The Author(s)). Thanks to these functionalities, TFLN holds significant promise for potential quantum photonics applications, such as enabling next generation quantum networks, advancing precision metrology, enhancing quantum computing architectures, and advancing toward near-deterministic single-photon sources.

medium, a high-quality photon-pair source, and fast, ultralow-loss switches.

In light of these considerations, an ideal integrated quantum platform would therefore feature low-loss propagation and coupling, high nonlinearity, ultra-fast electro-optic modulation, and compatibility with key components such as superconducting nanowire single-photon detectors (SNSPDs), quantum memories, and solid-state emitters, to achieve scalable quantum systems deployed in practical real-world environments. The platform must support precise control over the temporal and spectral profiles of photons to interface effectively with different platforms. It should also enable co-integration with cryogenic electronics, facilitating precise control and synchronization of quantum operations at low temperatures (<4 K). Moreover, the platform should support versatile quantum light sources capable of generating Fock states, squeezed light, and entangled photons. Together, these elements contribute to the development of a highly functional, scalable, and versatile quantum photonics platform, suitable for advanced applications in computing, communication, and sensing.

In terms of material choice, silicon (Si) and silicon nitride (SiN) have traditionally dominated integrated quantum photonics due to their well-established fabrication technology. However, emerging lithium niobate (LN)-based platforms (see Fig. 1), with loss metrics competitive with the best of other integrated photonics platforms, offer significant advantages across a broad spectral range. LN's strong second- and third-order nonlinearity enabling efficient nonlinear processes and its low-loss, ultra-fast modulation characteristics even at cryogenic temperatures through the Pockels effect are the capabilities that Si- and SiN-based platforms currently lack. In addition, LN is an excellent crystalline host for rare-earth ions,³⁰ making it one of the leading candidates for on-chip quantum memories.³¹ With the lack of single-photon detectors and sources, quantum memory can be potentially mitigated with heterogeneous integration (see Sec. 3).

In this paper, we provide an overview of the state-of-the-art TFLN quantum photonics and analyze its technological maturity and potential future challenges. Unlike previous overview papers on TFLN, focusing on photon-pair multiplexing,³² and more generic reviews of integrated quantum photonics,^{33,34} this paper will describe TFLN fabrication, heterogeneous integration with sources and detectors, nonlinear effects in TFLN, and quantum applications of TFLN devices.

The review is structured as follows. In Sec. 2, we consider the general properties of lithium niobate, types of waveguiding structures, and fabrication aspects of LN-integrated photonics and analyze the losses and linear elements performance achieved. Section 3 presents achievements in the heterogeneous integration of missing elements such as single-photon sources, detectors, and quantum memories. Section 4 contains an analysis of nonlinearity-based and system-level quantum applications followed by an outlook and description of challenges to overcome (Sec. 5).

2 Lithium Niobate Photonics

2.1 General Properties

Lithium niobate (LiNbO_3 , LN) is a dielectric and birefringent material. It has uniaxial crystal orientation, which is defined by the X and Y axes, known as the ordinary axes, and the Z axis, referred to as the extraordinary axis. LN exhibits a crystal structure, characterized by a 3m point group: mirror symmetry at 60-deg intervals and threefold rotational symmetry

along the c -axis, aligned with the Z -axis.³⁵ Its noncentrosymmetric and ferroelectric properties are responsible for its remarkable intrinsic characteristics,³⁶ making it an attractive material for various applications, e.g., nonlinear frequency conversion and electro-optic light modulation (Fig. 1). With a high Curie temperature close to its melting point (1210°C and 1257°C, respectively), a broad transmission range (from 350 nm to 5 μm), and good nonlinear coefficients ($r_{33} = 31$ pm/V, and $d_{33} = -25.2$ pm/V at 1064 nm),³⁷ LN is particularly compelling for photonic applications.³⁸

2.2 Types of LN Waveguides

Since the 1960s, bulk LN materials have been studied for electro-optic modulation and nonlinear optical processes. “Bulk LN” usually refers to a large, unstructured single-crystal material where surface effects are negligible. Single-crystal offers superior optical and electro-optic properties compared with polycrystalline forms rarely used in high-performance applications. Traditional bulk-LN waveguides and photonic devices were fabricated using proton exchange (PE) processes^{39–43} or metal in-diffusion, mostly Ti,^{44,45} to create low-index-contrast waveguides. The PE method replaces lithium cations (Li^+) with protons (H^+) by immersing LN in acidic solutions at elevated temperatures (150°C to 400°C) for several hours.^{39–43,46} Variations include annealed proton exchange (APE), reversed proton exchange (RPE), and soft proton exchange (SPE). SPE reduces proton concentration near the surface by adding lithium salts such as lithium benzoate to the acidic solution, achieving lower H–Li substitution ratios ($<10\%$).^{47–49} It preserves nonlinear coefficients and domain orientation without crystallographic phase transitions. Ti in-diffusion involves depositing Ti strips on the LN surface followed by thermal annealing, allowing Ti ion diffusion into the crystal and increasing the refractive index to form waveguides.

The refractive index change reaches a maximum of 0.1 for PE processes (and only for the extraordinary axis) and typically ranges between 0.001 and 0.04 for Ti in-diffusion. More details about PE processes and Ti in-diffusion can be found in previously published reviews.^{50,51} This low index contrast between bulk LN and the modified LN area results in weak mode confinement, reduced nonlinear efficiency, and large bending radii (on the order of millimeters), leading to significant footprints.⁵² Due to these limitations, combined with the challenges of etching LN and the strong competition from other photonic platforms, the LN platform was largely overlooked for several decades.

However, the emergence of thin-film lithium niobate (TFLN) wafers, in the 2000s, has positioned TFLN as one of the most promising candidates for integrated photonic platforms. The relatively high index of LN ($n_o \approx 2.2$ at 1550 nm) allows the formation of high-index contrast waveguides on top of low-index insulator materials, such as glass or sapphire. Significant progress in ion slicing and wafer bonding enabled the fabrication and commercialization of TFLN wafers.^{53,54} The “Smart Cut” process, originally developed for silicon-on-insulator (SOI) wafers,⁵⁵ was successfully adapted for LN.⁵⁶ This process involves four main steps:^{57,58} He^+ ion implantation, bonding to an insulator layer and substrate, controlled thermal splitting, and polishing. Companies now offer TFLN wafers with customizable LN thicknesses and various substrate materials (Si, sapphire, quartz, fused silica, ...), along with optional conductive

electrode layers (Au, Pt, Cr, ...) integrated above or below the isolation layer. The wafers could also be doped with alkaline earth and rare-earth elements to further enhance their properties.^{59–61}

The TFLN platform is now widely used for integrating high-performance devices across a broad range of applications, including high-speed electro-optic modulation,^{62–68} electro-optic frequency conversion,⁶⁹ acousto-optic modulators,^{70,71} acoustic wave filters,⁷² optical^{73–77} and Kerr frequency comb generation,^{78–82} nonlinear wavelength converters,^{83–85} entangled photon-pair sources,⁸⁶ transduction,^{87–91} and photonic quantum technologies.^{32,33} The terms “LNOI” and “TFLN” are commonly used interchangeably to describe an integrated platform comprising a submicron-thick LN film bonded to a SiO₂ layer on a Si substrate. In this work, we chose the term “TFLN” to refer to this specific platform, unless otherwise stated for cases involving thicker films and/or different substrate materials. Table 1 summarizes the most important performance metrics for various implementations of LN, including bulk and TFLN.

2.3 Integration of Photonic Devices into the TFLN Platform

Despite its attractive intrinsic properties, lithium niobate has historically faced integration challenges due to difficulties in the etching process. Today, dry etching is the most common method for achieving low-loss LN devices. It is widely used in the semiconductor industry due to its ability to produce anisotropic profiles and provide precise control over etching depth. However, dry etching of lithium niobate has been a longstanding challenge primarily due to redeposition phenomena resulting in rough and nonvertical sidewalls. A detailed discussion of mask selection and various gases used in the dry etching process is provided below.

2.3.1 Resist and hard masks

Device fabrication on TFLN requires an intermediate mask layer for pattern transfer through lithography. For large structures, typically with critical dimensions greater than 1 μm , UV resists such as S1828,⁹⁸ SU-8,⁹⁹ and SPR-955 CM-0.9¹⁰⁰ are used, whereas sub-micrometer features require deep-UV or e-beam lithography with positive (CSAR 62,¹⁰¹ ZEP-520A,^{102–105} and PMMA^{72,106}) or negative (HSQ^{62,64} and Ma-N^{107,108}) resists. Hard masks enhance etching selectivity and depth but require additional patterning steps. Various materials have been investigated including amorphous-silicon,^{109,110} silicon oxide,¹¹¹ nickel,^{98,112,113} diamond-like carbon (DLC),¹¹⁴ silver,¹¹² and chromium^{96,115–118} with its alloys.^{52,119} Notable results include DLC-masked etching achieving 4 dB/m losses¹¹⁴ and

chromium-masked waveguides with 2 dB/m propagation losses.¹²⁰

2.3.2 Etching gases

Fluorine gases (SF₆,^{42,43} CHF₃,¹²¹ or CF₄¹²²) can etch lithium niobate but form a nonvolatile LiF material. LiF redeposition creates rough and sloped sidewalls and reduces etching rates. Proton exchange (PE) minimizes LiF formation by replacing Li⁺ with H⁺ but degrades electro-optical properties.^{52,123} Enhanced sputtering using Ar gas with SF₆,^{43,52,124,125} CH₃F,^{126,127} C₃F₈ or He gas with C₄F₈^{128,129} or CF₄⁴² has been tested. CH₃F/Ar mixtures on PE-LN achieved nearly vertical walls.¹²³ Chlorine-based etching using Cl₂,^{130,131} BCl₃, or Ar/Cl₂ mixtures produced smooth sidewalls with 80 deg angles.¹¹⁰

Nevertheless, the use of pure physical etching with Ar gas has emerged as the most adopted technique for integrating devices into TFLN platforms.^{62–64,72,96,101,103,104,107–109,114,115,132–134} The influence of the etching parameters has been studied to optimize dry etching.^{98,135} Kaufman et al.¹³⁶ carried out a detailed study to achieve a redeposition-free regime and strongly emphasized the importance of chamber maintenance and conditioning to achieve good repeatability and consistency in the results. Post-etching steps are also used to optimize the device integration. Soaking in basic mixtures or solutions such as (NH₄OH:H₂O₂:H₂O) at 70°C⁹⁸ or KOH^{136,137} effectively removes the redeposited nanoparticles induced by the Ar physical etching. Post-etch annealing in O₂ or air atmospheres for several hours further reduces optical losses by restoring TFLN optical properties to almost bulk LN levels. It also improves the intrinsic Q-factors by up to fourfold.^{28,95,120,138–141} Zhang et al.¹⁴² achieved low propagation losses as 2.7 dB/m at 1550 nm and microring resonators with quality factors as 10⁷. Zhu et al.⁹⁵ achieved an intrinsic Q-factor of 29 million for TFLN microresonators corresponding to a record-low propagation loss of <1.3 dB/m.

2.3.3 Other integration methods

Other fabrication methods have been investigated to integrate devices into the TFLN platform. For instance, wet etching offers reproducible and scalable fabrication processes. The methods using HF and HNO₃^{41,128} with Cr masks and NH₄OH, H₂O₂, and H₂O mixture^{143,144} with SiO₂ masks have been studied. However, this method tends to produce isotropic profiles, leading to under-etching beneath the mask. This creates design limitations as closely spaced trenches may merge if placed too close to each other.

Ion beam-enhanced etching (IBEE) is another used method.^{145,146} In this process, after mask etching, the exposed LN surface is irradiated with Ar⁺ ions of varying energies,

Table 1 Comparison of lithium niobate-based platforms.

Platform	Δn	Mode field area	α_{prop} , dB/m	Bending radius, μm
Bulk	0	N/A	0.2 ²⁸	N/A
PE (inc. APE, RPE, SPE)	0.1 ⁹²	>1.4 μm^2 ²⁹²	20	>500 μm ⁹²
Ti-diffused	<0.01 ⁹³	>5 μm^2	50 ⁹³	~1 mm ⁹⁴
TFLN	0.7	$\lesssim 1 \mu\text{m}^2$	1.3 ⁹⁵ (etch) 0.34 ⁹⁷ (CMP)	<100 μm ⁹⁶

followed by wet etching to remove the irradiated LN material while preserving the masked regions. The focused ion beam (FIB) technique has also been applied to integrated photonic structures in LN crystals or into the TFLN platform.^{57,147–152} A high-energy ion beam is focused and aimed at the sample, removing atoms from the surface through sputtering.

Mechanical techniques such as optical grade dicing have also been investigated but are limited to low-scalability fabrication of regular structures with feature sizes larger than $1\ \mu\text{m}$.^{153–156} Chemical-mechanical polishing (CMP) has been used for the fabrication of LN waveguides and microring resonators.^{97,141,157–163} The process involves chromium mask patterning via laser ablation, CMP polishing, wet chromium removal, and optional secondary CMP for surface smoothing. With this method, Gao et al.⁹⁷ achieved microring resonators with intrinsic Q -factors of 1.08×10^8 , corresponding to 0.34 dB/m propagation losses. Post-etching CMP can also be used to remove LiF redeposition and polish sidewalls after fluorine dry etching.^{164,165} Wolf et al.¹⁶⁵ combined Ar + CHF₃ etching with CMP to fabricate microring resonators achieving 4 dB/m losses at 980 nm for the TE mode.

2.4 Coupling Interfaces with Fibers

Fiber-to-chip couplers are essential for interfacing between optical fibers and TFLN photonic circuits. Efficient coupling to TFLN chips relies on two main approaches: edge couplers and grating couplers (GCs).^{166,167}

The principle of edge couplers involves tapering the waveguide dimensions for spot size conversion (SSC) to expand its mode field diameter (MFD) to match the fiber mode. Edge couplers interface optical fibers with LN waveguides at the chip edge, typically providing a large coupling bandwidth. Coupling losses of 1.32 dB for TE modes and 1.88 dB for TM modes with broad bandwidth by tapering the coupled fiber have been reported.¹⁶⁸ However, edge couplers typically require precise alignment and polished facets, making them unsuitable for wafer-scale testing. Detailed design principles of polarization-insensitive edge couplers have been discussed in Ref. 169. Designing efficient edge couplers typically involves a tapering strategy of the TFLN waveguide, either horizontally or vertically. Tapering TFLN waveguide horizontally is typically applied, which could be well defined by a lithography process. Different SSC designs have been studied to achieve efficient coupling with fibers. A TFLN bilayer inverse taper-based SSC reduced coupling losses to 1.7 dB per facet at 1550 nm.¹⁷⁰ A tri-layer edge coupler design on the TFLN platform with a SiN assisting waveguide achieved 0.75 dB/facet at 1550 nm and 0.64 dB/facet at ~ 1610 nm, though alignment and fabrication deviations remain critical, with ~ 0.15 dB loss from SiN-to-LN mode conversion.¹⁷¹ Combining silica-based planar lightwave circuits with an LN bilayer taper yielded coupling losses of 0.11 dB (TE) and 0.128 dB (TM) with excellent scalability.¹⁷² A bilayer taper structure with narrow tips (<300 nm) achieved 0.54 dB (TE) and 0.59 dB (TM) at $1.55\ \mu\text{m}$ using an ultra-high aperture fiber.¹⁷³ To release the alignment sensitivity in the fabrication process, a crossed bilayer taper was proposed and demonstrated, exhibiting low coupling loss of 0.29 and 0.24 dB for TE and TM modes to a high-numerical-aperture fiber (HNAF), respectively, with excellent fabrication tolerance.¹⁷⁴ Tapering vertically is typically requiring grayscale lithography.¹⁷⁵ Jia et al.¹⁷⁶ proposed and demonstrated a novel vertical tapering process controlled by CMP with a chromium mask. A coupling loss of

0.92 dB to HNAF for the TE₀ mode was reported, assisted with a SiON waveguide.¹⁷⁶ The same group further simplified the design without SiON and with SiO₂ cladding, which leads to a coupling loss of 1.43 dB for the TE₀ mode with the lensed fiber.¹⁷⁷ These innovations demonstrate significant progress in improving coupling efficiency and scalability for TFLN edge couplers.

Grating couplers (GCs), in contrast to edge couplers, enable vertical coupling via diffractive structures, offering the possibility of wafer-scale testing. However, TFLN GCs face challenges from LN's anisotropy and etching limitations, causing nonvertical sidewalls that restrict pitch size and fill factor. GCs typically provide coupling bandwidths of tens of nanometers, which would restrict their applications, e.g., in wide-band optical communications. TFLN-based GCs are categorized into one-dimensional (1D) and two-dimensional (2D) designs. 1D GCs with optimized periodic structures achieve efficient coupling to fiber. Yet, limitations such as scattering losses, substrate leakage loss, and mode mismatch result in coupling losses between 3 and 10 dB.^{178–183} Mitigation strategies include buried metal reflectors to redirect downward-scattered light back into the grating, enhancing efficiency. In addition, techniques such as apodized gratings, angled-etched gratings, and multilayer structures offer improved coupling efficiency.^{178,180,181,184–190} For example, an apodized 1D GC on Z-cut TFLN with a gold reflector achieved losses as low as 1.42 dB.¹⁸⁴ Furthermore, hybrid structures with high-index materials, such as amorphous silicon, improve grating sharpness, achieving efficiencies up to 50% per coupler with a 1-dB bandwidth of 55 nm.¹⁸⁰ In comparison, 2D GCs enable functionalities such as polarization multiplexing but exhibit higher losses, such as -5.13 dB for P-polarized light and -7.6 dB for S-polarized light, lagging behind optimized 1D designs.¹⁹¹

1D GCs are polarization-sensitive in general. Efforts to reduce polarization sensitivity include metal-based 1D GCs on X-cut TFLN. Using plasmonic modes, these designs align diffractive angles for TE and TM polarized light, reducing polarization-dependent losses to less than 0.69 dB across a 44 nm wavelength range. This approach achieved losses of 3.56 dB for TE modes and 4.08 dB for TM modes. Further improvements, such as cavity-assisted gratings with top metal mirrors, have reduced losses to 0.89 dB,¹⁸⁸ though these methods add fabrication complexity.

2.5 Switches and Modulators

Optical switching and modulation can be realized easily by the thermo-optic effect. A micro-heater is typically placed either on top or¹⁹² aside [see Figs. 2(a) and 2(b)]^{193,194} the LN waveguide with sufficient distance to reduce the metallic absorption loss. Applying a voltage or current on the microheater generates heat, which changes the refractive index of the LN waveguide. Due to the slow heat dissipation, microheater-based phase shifters have typical bandwidths of tens of kHz.¹⁹³ The thermo-optic coefficient of n_e and n_o of lithium niobate is around $3.6 \times 10^{-5}\text{K}^{-1}$ and $4.2 \times 10^{-5}\text{K}^{-1}$,¹⁹⁵ respectively. A typical power consumption, P_π (half-wave power consumption), of 44 mW¹⁹⁶ and 760 mW,¹⁹⁴ is achieved for microheaters designed on top and aside LN waveguides, respectively. To reduce the power consumption, the under-cutting technique can be applied leading to a power consumption improvement by a factor of 20 times.^{192,193} However, it also greatly reduces the switching speed to sub-kHz, due to much slower thermal dissipation. To increase

the speed of the microheater, the driving signal can be pre-engineered to result in a fast phase rising time of only a sub-microsecond [see Fig. 2(c)].¹⁹⁶

If more than GHz bandwidth is required, the Pockels effect of LN can be utilized. A more general description of the Pockels effect and basics of electro-optic modulation can be found in classical books, e.g., written by Boyd.¹⁹⁷ The main advantage of the Pockels effect is pure phase modulation with ultra-fast response time (picoseconds and less). For a simple introduction, we would like to consider an X-cut LN phase shifter [see Fig. 2(d)]. To understand the principle, one considers a device where refractive index changes are induced by an external electric field E_z applied along the crystalline Z-axis. The change of the refractive index can be described as $\Delta n = -\frac{1}{2}n_e^3r_{33}E_z$, where n_e is the extraordinary refractive index value and $r_{33} = 30.9$ pm/V is the zzz component of the electro-optic tensor of LN. Induced phase shift in this case can be expressed as $\Delta\phi = -\frac{1}{2}n_e^3r_{33}E_zk_0L = -\pi\frac{V}{V_\pi}$. Here, we expressed our field $E_z = \frac{V}{d}$ in terms of applied voltage V and electrodes gap d and also introduced half-wavelength voltage V_π . This effect allows pure phase E/O modulation desirable for quantum photonics. The pure-phase modulator can be incorporated in a Mach–Zehnder interferometer (MZI) leading to high-speed and high-fidelity switches with ultralow loss, which is crucial for demultiplexing of, e.g., quantum dot-based single-photon sources,¹⁹⁸ multiplexing of nonlinear photon-pair sources,^{199,200} and universal quantum computing.²⁰¹ Furthermore, cascading TFLN switches not only allows for on-chip reconfiguration of the optical propagation mode,²⁰² which would be very useful in on-chip mode-encoded entanglement applications,²⁰³ but also allows for quick manipulation of the polarization state of light.²⁰⁴ Advanced in-phase/quadrature (IQ) modulators can be constructed by two Mach–Zehnder modulators (MZMs) [see Fig. 2(e)],²⁰⁵ as is widely used in continuous variable (CV) quantum key distributions (QKD). There are a few key metrics that need to be taken into consideration for evaluating the performance of the LN switches and modulators, including the optical insertion loss, the $V_\pi L$ product, which describes the modulation efficiency of the device, the extinction ratio, which is important for high fidelity preparation and manipulation of quantum states, and the E/O bandwidth, which limits the highest operation rate of the device. To design an effective high-bandwidth modulator, overlap integral and phase-matching conditions between RF and optical modes should be taken into account. In addition, a relative position between electrodes and optical field is important for propagation loss management. A deeper analysis of high-speed TFLN modulators and EO devices and specific of its design can be found in two fundamental reviews from Harvard University.^{67,206}

The traditional E/O LN modulators are based on Ti-diffused or PE waveguides, as discussed in Sec. 2. The weak light confinement requires a large distance between the radio frequency (RF) electrodes, leading to low modulation efficiency (typically >10 V · cm). A sub-1 V V_π (half-wave voltage) thus results in a long device of at least 10 cm. Moreover, due to RF signal loss in the long RF electrodes, the bandwidth is typically limited to less than 40 GHz. The TFLN provides high light confinement that eliminates the bottlenecks of the traditional LN modulators and switches in terms of footprint, modulation efficiency, and bandwidth, with a typical modulation efficiency around 2 to

3 V · cm. Significant progress has been achieved in high-performance E/O modulation, such as TFLN modulators with CMOS-level driving voltages being reported, relying on the regular coplanar RF waveguide scheme.⁶² The regular coplanar RF electrode design still suffers from phase mismatch, which is greatly improved by capacity-loaded traveling-wave (TW) electrodes and employing quartz substrate. A 2 cm long EO modulator with a V_π of only 1.3 V is reported.²⁰⁷ The high-performance LN phase modulators have naturally led to advanced in-phase/quadrature (IQ) modulators. High bandwidths of 67 GHz and driving voltages of 3.1 V²⁰⁵ have been reported. Recently, dual-polarization IQ modulators have also been reported.²⁰⁸ Though those demonstrations are mainly focused on high-capacity optical communications, they can be widely used in CV-QKD systems²⁰⁹ to achieve Gaussian modulation. The modulation efficiency of TFLN modulators still leads to a device length of around 1 cm. To further reduce the footprint, folded MZMs incorporating meander traveling wave electrodes and optical waveguides have been reported,^{210,211} with the price of more complicated electrical driving circuits and complex packaging.

3 Heterogeneous Integration on TFLN

TFLN does not provide a native solution for generating and detecting light, but this may be mitigated by the use of hybrid integration techniques.

3.1 Integration of Single-Photon Emitters on TFLN

The first demonstration of quantum dot integration on the TFLN platform was demonstrated²¹² in 2018 by Aghaieimbodi et al. The authors prepared a sample of InAs dots in an InP matrix with a mode converter to couple into the TFLN waveguide. Using the “pick-and-place” approach (which today may be referred to as “microtransfer printing”²¹³), an InP nanobeam with a Bragg reflector and a 5 μ m adiabatic taper was mounted on a TFLN waveguide. The coupling efficiency from the taper to the waveguide was 40.1%, which can be improved by using a longer taper (see the Supplementary Material of Ref. 212) up to 60% (−2.2 dB). The total collection efficiency, considering the QD emission efficiency (85%) and grating coupler efficiency (26.7%), was found to be 9%. The experimentally measured collection efficiency of 2.2% differs from the theoretically expected value, which can be explained by fabrication imperfections and residual misalignment between the nanobeam and the waveguide.

This approach was developed further by Wei and Kesse in their work²¹⁴ in 2021 [see Fig. 3(a)]. They transferred a similar structure with the Bragg reflector, adiabatic taper, and InAs/GaAs QD emitter on the TFLN circuit with tunable directional couplers for path multiplexing of emitted single photons. Although 50 μ m long taper demonstrated the near-unity transmission (>0.9 , corresponding to <-0.46 dB loss), the overall source efficiency reached 42%, primarily due to low coupling between QD and GaAs waveguide. These results demonstrate the capability of almost lossless transition of the single photon to the TFLN chip. The main limitation still present is a design for compact structures to couple light from the emitter to the hosting platform waveguide. From our perspective, this issue can be addressed, and the heterogeneous integration approach appears promising—offering better scalability and environmental stability compared with multiple fiber interfaces between the

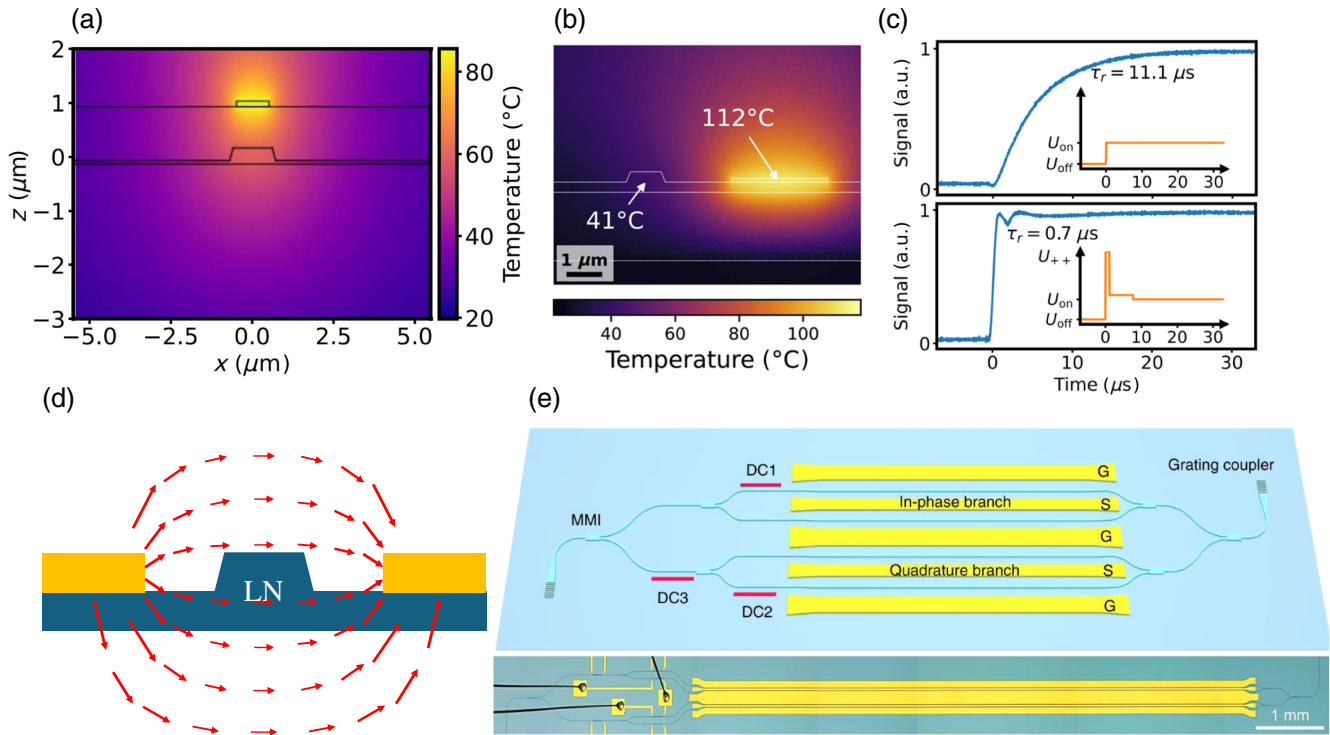


Fig. 2 Light modulation based on micro-heaters and the Pockels effect. (a) Microheater-based phase shifter on top of LN. Adapted with permission from Ref. 196 © 2024 The Author(s). (b) Microheater-based phase shifter aside the LN waveguide. Adapted with permission from Ref. 194 © 2022 Optica Publishing Group. (c) Synthesizing the electrical driving signal enables faster thermal tuning. Adapted with permission from Ref. 196 © 2024 The Author(s). (d) Applying an electrical field results in high-speed phase modulation of an LN waveguide. (e) Advanced IQ modulator based on high-speed phase modulation. Adapted with permission from Ref. 205 © 2020 The Author(s).

emitter and processing chip while also avoiding the nonuniform behavior of connecting fibers.

In the last couple of years, a group from KTH demonstrated QD emission tuning using the piezoelectric effect in lithium niobate for InAsP QD in an InP nanowire [see Fig. 3(b)],^{29,215}

thus introducing an additional degree of freedom for single-photon sources.

Alternatively, instead of using QD emitters, White et al.²¹⁶ demonstrated the integration of a WSe₂ monolayer-based single emitter on a Ti-diffused LN coupler.

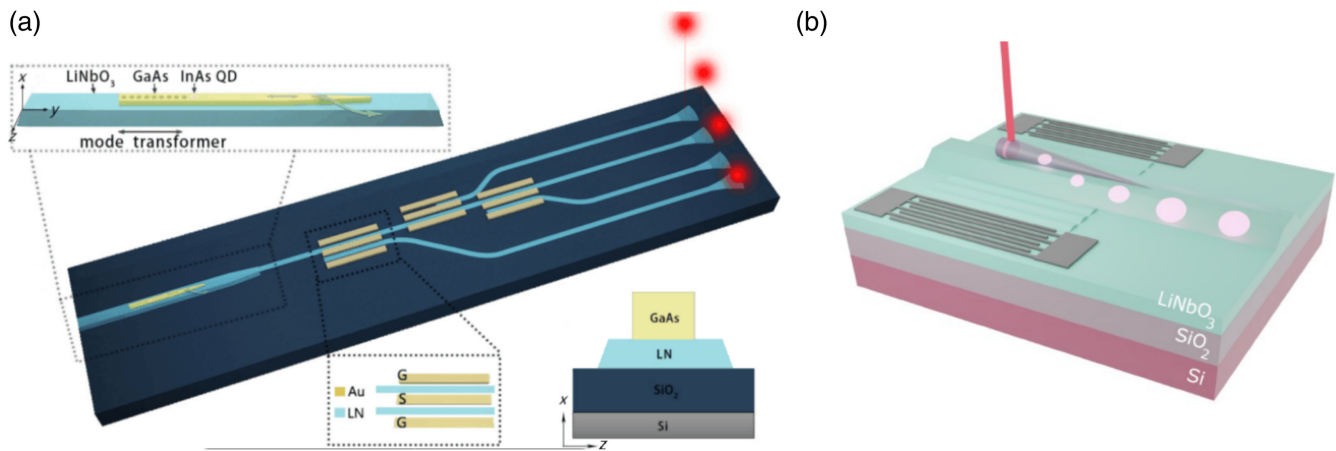


Fig. 3 Integration of the single-photon emitters. (a) InAs/GaAs. Adapted with permission from Ref. 214 © 2021 The Author(s). (b) InAsP/InP nanowire integration with piezoelectric transducer. Adapted with permission from Ref. 29 © 2023 The Author(s).

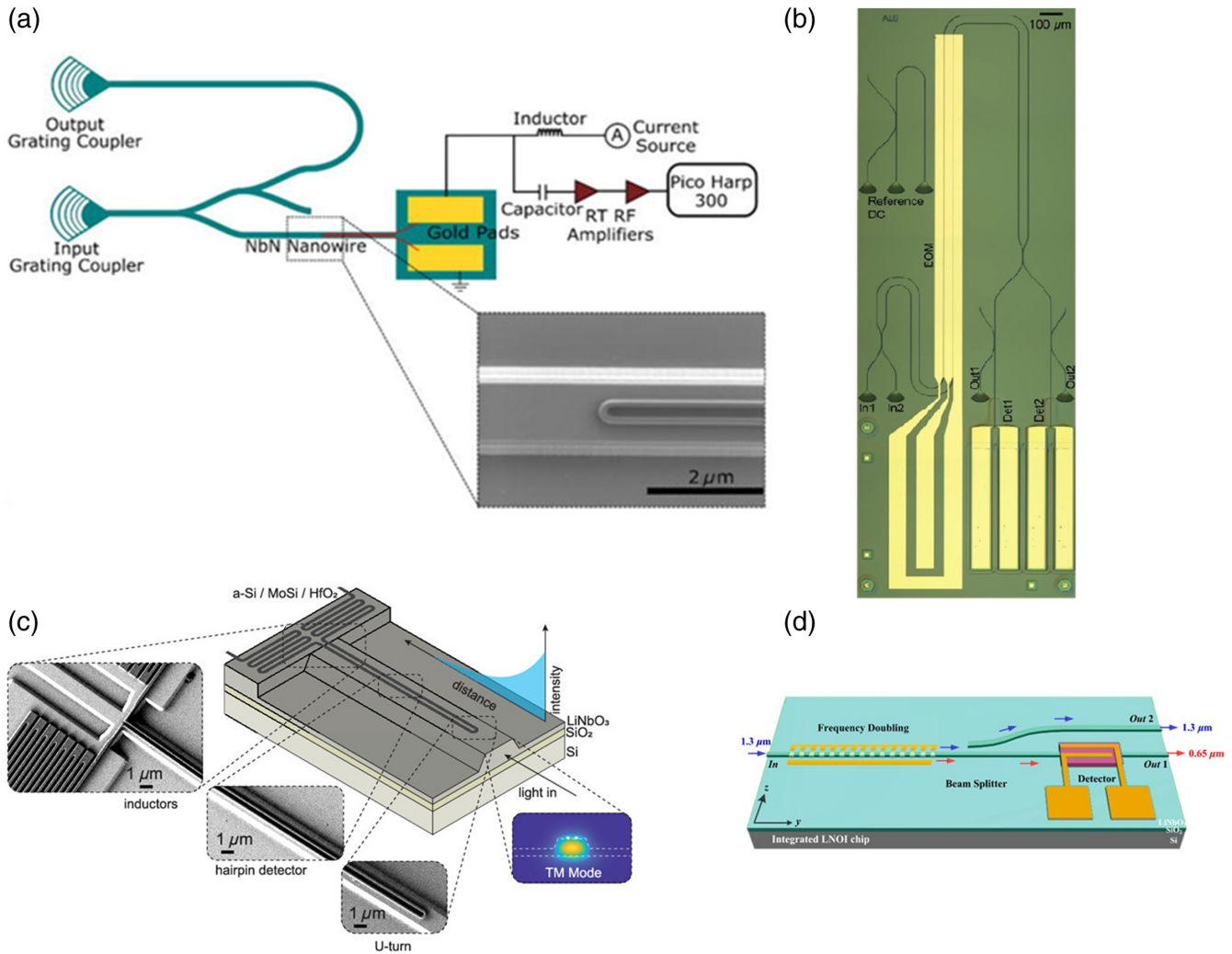


Fig. 4 SNSPD integration. (a) NbN nanowire integrated on TFLN. Adapted with permission from Ref. 218 © 2020 AIP Publishing. (b) NbTiN-based SNSPDs on TFLN. Adapted with permission from Ref. 219 © 2021 The Author(s). (c) MoSi nanowire integrated on TFLN. Adapted with permission from Ref. 220 © 2024 American Chemical Society. (d) α -Si detector on TFLN. Adapted with permission from Ref. 221 © 2024 American Chemical Society.

3.2 Integration of Detectors on TFLN

To integrate the detectors on TFLN, the SNSPD is the best option in terms of detection efficiency and jitter. The description, working principle, and progress on a different material platform can be found in a freshly published SNSPD review.²¹⁷ The integration of a NbN nanowire on a TFLN waveguide with 46% on-chip detection efficiency and 32 ps timing jitter was shown by Sayem et al.²¹⁸ [see Fig. 4(a)]. Besides NbN, NbTiN can also be used as material for detectors, and the integration of such nanowire on a TFLN waveguide was reported in Refs. 219 and 222 [see Fig. 4(b)]. Lomonte et al.²¹⁹ demonstrated NbTiN SNSPD with 27% on-chip detection efficiency and 17 ps jitter. Recently, Prencipe et al.²²² demonstrated a wavelength-sensitive SNSPD integrated into TFLN, which potentially opens the path to wavelength detectors. In 2024, Colangelo et al.²²⁰ demonstrated 50% on-chip detection efficiency with 82 ps jitter at 1550 nm utilizing the integration of a MoSi nanowire on TFLN [see Fig. 4(c)]. In the same year, He et al.²²¹ demonstrated

a heterogeneous integration of an amorphous silicon detector (α -Si) on TFLN [see Fig. 4(d)], utilizing a preceding on-chip upconversion, which, despite low external efficiency, demonstrated potential single-photon detection even at room temperature.

3.3 Quantum Memory and Storage

For storage applications, the so-called photon buffers at the single-photon level on a TFLN chip functioning at room temperature were recently demonstrated²²³ [see Fig. 5(a)]. They exhibit low-loss characteristics, with short-term storage capabilities of up to 1.4 and 1.8 ns and time resolutions of 100 and 200 ps, respectively. Such short, controllable storage media have been exploited to temporally multiplex single-photon sources.²⁴ Moreover, these buffers are expected to enable a variety of novel on-chip photonic applications, including the generation of multiphoton entangled states²²⁶ and the multiplexing of probabilistic gates.²²⁷ Operating in the picosecond regime is particularly

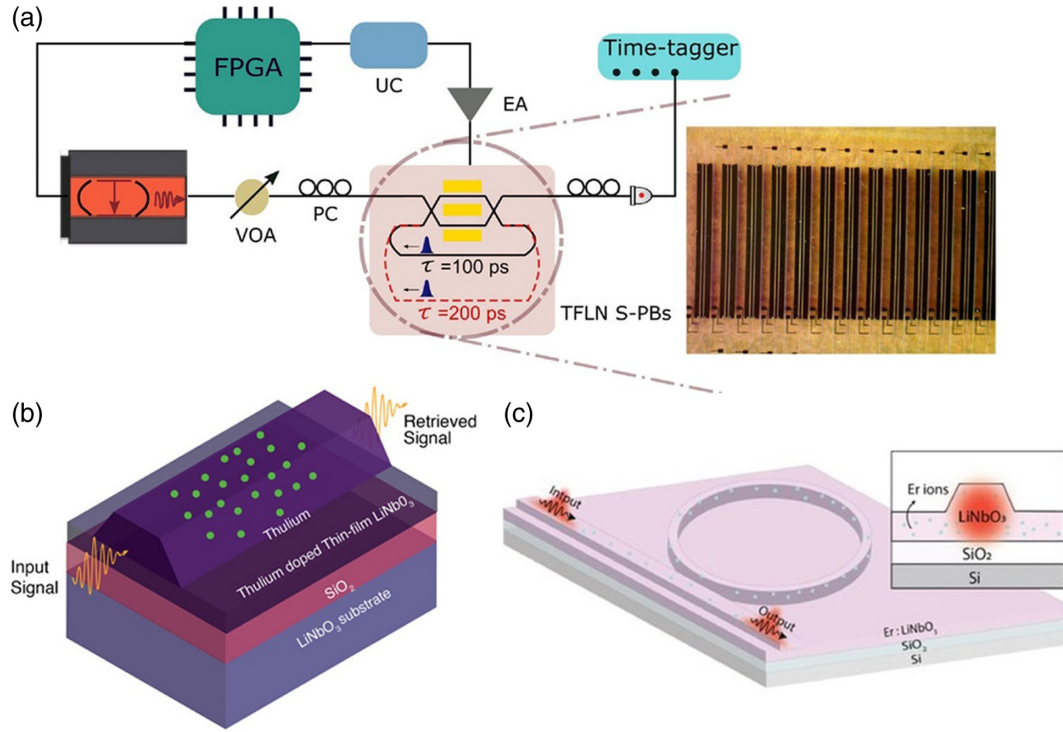


Fig. 5 Quantum memory and storage. (a) TFLN-based single-photon buffer. Adapted with permission from Ref. 223 © 2023 The Author(s). (b) A quantum memory on TFLN doped with thulium rare-earth ions. Adapted with permission from Ref. 224 © 2023 American Chemical Society. (c) Er-doped TFLN ring resonator cavity for telecom-compatible quantum memory. Adapted with permission from Ref. 225 © 2014 SPIE.

advantageous as it allows multiplexing within the recovery time of commercial SNSPDs, typically on the order of tens of nanoseconds, required for heralding. This enables many probabilistic events, such as a photon-pair generation, to be attempted within that time window. On the other hand, for long-term storage, LN is an outstanding crystalline host for rare-earth ions, positioning it as one of the best candidates for on-chip quantum memories, with coherence times on the order of microsecond for achieving sufficient storage times and high-efficiency storage for multiple temporal modes.³⁰ A recent demonstration reported a compact integrated storage device based on TFLN waveguide doped with thulium rare-earth ions, with a broad storage bandwidth exceeding 100 MHz and optical storage time as long as 250 ns²²⁴ [see Fig. 5(b)]. In addition, erbium (Er)-doped TFLN waveguides utilizing a nanophotonic cavity have recently been suggested to pave the way for a quantum memory at telecom-compatible wavelengths²²⁵ [see Fig. 5(c)].

4 Quantum Applications of TFLN

4.1 Nonlinear Quantum Applications

4.1.1 Nonlinear susceptibilities of lithium niobate

The presence of the second- and third-order nonlinear susceptibility combined with high field confinement makes TFLN a very useful material for the integration of various efficient active and nonlinear devices.²²⁸ The second-order nonlinearity in lithium niobate appears in terms of $d_{33} = -25.2$ pm/V, $d_{31} = -4.6$ pm/V, and $d_{22} = 2.1$ pm/V; all values are given at 1064 nm.^{37,229} It is important to keep in mind that these values

also have a wavelength dispersion and can vary up to 25%, e.g., the value of $d_{33} = -20.2$ pm/V at 1550 nm measured,²³⁰ which differs from the standard value of -25 to 27 pm/V^{37,229,231} widely referred to in the literature. Although the high second-order nonlinear coefficient dominates, the third-order nonlinearity of lithium niobate at 1550 nm ($n_2 = 1.8 \times 10^{-19}$ m²/W)²⁰⁶ is also useful, being on a level comparable with that of silicon nitride ($n_2 = 2.5 \times 10^{-19}$ m²/W).

The aforementioned properties give access to a wide set of the nonlinear processes (see Fig. 6), which can be used for frequency conversion [second harmonics generation (SHG), sum-frequency generation (SFG), difference frequency generation (DFG), and Bragg-scattering four-wave mixing (BS-FWM)], pair generation [four-wave mixing (FWM), degenerate FWM (DFWM), and parametric down conversion (PDC)], and squeezed state generation (DFG).

The nonlinear optical processes require phase matching to be effective. It can be seen from a simple consideration of coupled-mode equations in slowly varying envelope approximation for the SFG process $\omega_3 = \omega_1 + \omega_2$,

$$\partial_z A_1(z) = -i\kappa_1 A_3(z) A_2^*(z) \exp(-i\Delta k z),$$

$$\partial_z A_2(z) = -i\kappa_2 A_3(z) A_1^*(z) \exp(-i\Delta k z),$$

$$\partial_z A_3(z) = -i\kappa_3 A_1(z) A_2(z) \exp(i\Delta k z),$$

where, $A_i, i \in \{1, 2, 3\}$ is the envelope function, κ_i is the wave coupling coefficients, $\Delta k = k_3 - k_2 - k_1$ is the wavenumber

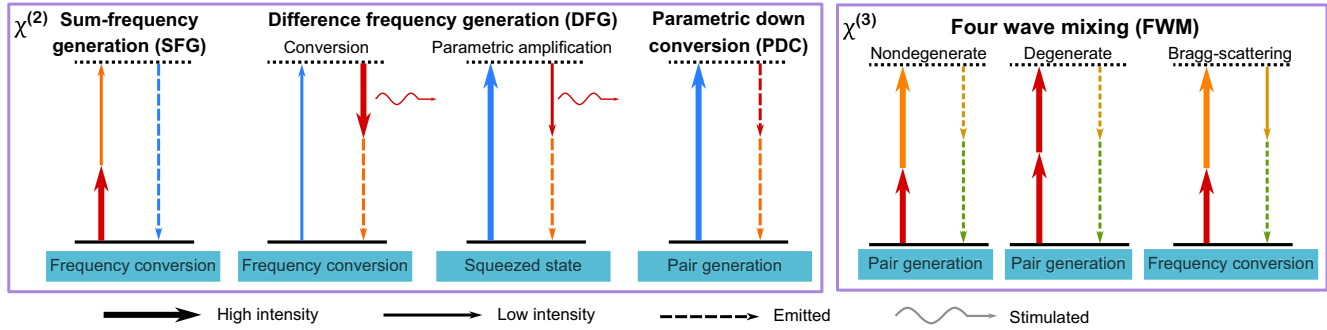


Fig. 6 Nonlinear processes of the second and third orders with their potential quantum applications.

mismatch, and $\exp(i\Delta kz)$ term is responsible for phase-matching. The oscillatory nature of this term leads to periodic changes in the energy transfer direction in the case of $\Delta k \neq 0$, significantly reducing the efficiency of the generation. More insights and deeper consideration of second-order nonlinear processes can be found in classical books¹⁹⁷ or in fundamental tutorial papers.^{206,232}

Thus, some measures should be applied to compensate for this mismatch. In integrated photonics, this can be achieved via dispersion and/or nonlinearity engineering. Dispersion engineering helps to design the waveguide cross-section to compensate for the chromatic dispersion of the material with waveguide and modal dispersion.

The spontaneous polarization of lithium niobate can be flipped by applying an external electric field due to its ferroelectric nature. This leads to changing the sign of the second-order nonlinear coefficient, making it possible to obtain quasi-phase matching (QPM).²³³ In QPM, the phase mismatch $\Delta k = k_3 - k_2 - k_1 - \frac{2\pi m}{\Lambda} = 0$ is compensated by a crystal momentum component of the periodic structure with period Λ of polarization poled regions.

These nonlinear effects may find applications in spectral conversion and manipulation of the shape of a wave function at the single-photon level or even for generating quantum light. A broader discussion of nonlinear processes, including other material platforms, can be found in the review of Dutt et al.²³⁴

4.1.2 Quantum frequency conversion

To link different technologies, optimized for different wavelength regions, such as quantum memories based on Rb atoms operating at 780 nm,²³⁵ single-photon sources operating at 900 to 980 nm,²³⁶ and optical fibers with the lowest losses around 1550 nm,^{237,238} it may be highly useful to be able to effectively shift the frequency of the photons. The whole idea of quantum frequency conversion (QFC) is that an incoming single photon can be converted to another frequency range without a change in its statistics. This is challenging as the nonlinear media with the presence of the strong (classical) pump could easily influence the quantum state, if special care is not taken. An early concept for QFC was proposed²³⁹ and experimentally observed²⁴⁰ by Kumar and Huang relying on the SFG in periodically poled potassium titanyl phosphate (PPKTP) crystals. Ou²³⁷ later did a theoretical study of potential processes for quantum state conversion, including three- and four-wave mixing and Raman and Brillouin scattering. QFC has been seen to preserve entanglement,²⁴¹ photon number purity, and indistinguishability.²⁴²

QFC has been experimentally demonstrated using PPLN waveguides under cryogenic temperatures.²⁴³ The most important QFC experiments demonstrated in LN are summarized in Table 2 with a comparison in terms of converter efficiency (including propagation losses) listed chronologically. Some results include coupling losses due to lack of information to subtract it, and these results are listed with a star symbol.

One of the first applications of QFC was for single-photon detection using Si detectors, operating optimally in the near visible range, utilizing a preceding upconversion. This concept has been demonstrated in PPLN crystals,^{244,245} RPE waveguides,²⁴⁶ and TFLN²⁵⁸ with high (70% to 90%) internal conversion efficiency and around 40% detection efficiency for 1550 and 1320 nm²⁴⁶ signals. Pelc et al.²⁴⁸ demonstrated 86% internal conversion efficiency and 37% total detection efficiency at 1.55 μm , which is considered to be 94% and 45%, respectively. In addition, it was shown that Raman scattering and spontaneous PDC (SPDC) were the two main noise sources.²⁶⁰

For a long time, demonstrations of QFC used upconversion via SFG. However, frequency converters for downconversion based on DFG became interesting alongside the development of nonclassical sources, such as nitrogen vacancy (NV) centers in diamond,²³⁸ quantum dots,²⁵⁷ and Rb memory storage,²⁶¹ to translate generated single photons to the longer wavelength where the optical fiber has minimum losses. One of the challenges for DFG realization is the requirement of a high-power pump in the IR and mid-IR ranges, to provide low noise conversion.

From Table 2, it can be seen that only in one recent paper,²⁵⁸ a TFLN-based converter has been used. The main challenges for end-to-end device performance are the fiber-to-chip loss and propagation loss in the waveguide. This draws special attention to the need for optimized design of fiber-to-chip couplers and fabrication optimization.

In quantum information processing and computing, the indistinguishability of generated and processed photons is crucial for an overall system performance. A key scalability challenge arises from fabrication-induced nonuniformity among quantum emitters. Thus, the ability to change not only central frequency but also the temporal and spectral shape of a single-photon wave package becomes important. Nonlinear conversion processes are one possible approach to achieving this control, contributing to more precise quantum interfaces and improved scalability. Using a PPLN waveguide for frequency conversion, Takesue²⁶² demonstrated the erasure of frequency distinguishability while preserving the Fourier-transform limited characteristics of the

Table 2 Summary of quantum frequency conversion experiments and their application.^a

Process	λ_{in} , nm	λ_{out} , nm	Material	η_{device}	Application (Ref.)
SFG	1548	631	Bulk PPLN	~90%	Upconversion detection ²⁴⁴
SFG	1550	631	Bulk PPLN	80% $\pm$ 15%	Upconversion detection ²⁴⁵
SFG	1550	713	PPLN RPE	46% ^b	Upconversion detection ²⁴⁶
SFG	1320	713	PPLN RPE	40% ^b	Upconversion detection ²⁴⁶
SFG	1312	710	PPLN WG	75% $\pm$ 1%	Quantum interface ²⁴⁷
SFG	1554	834	PPLN RPE	86%	Upconversion detection ²⁴⁸
DFG	780	1522	Zn:PPLN WG	62%	Quantum interface ²⁴⁹
SFG	980	600	MgO:PPLN WG	>70%	QFC ²⁴²
DFG	711	1313	Zn:PPLN WG	>64%	QFC ²⁵⁰
DFG	910	1560	PPLN RPE	80%	Quantum interface ²⁵¹
DFG	780	1552	MgO:PPLN WG	32.4%	Quantum interface ²³⁵
DFG	637	1587	Zn:PPLN WG	71%	Quantum interface ²³⁸
DFG	904	1557	MgO:PPLN WG	31.1% ^b	QFC ²⁵²
SFG	1545	550	PPLN TD	61.5%	Quantum interface ²⁵³
DFG	606	1552	PPLN WG	62%	Quantum interface ²⁵⁴
DFG	795	1342	PPLN RPE	70%	Quantum interface ²⁵⁵
DFG	942	1550	PPLN WG	56.7%	Quantum interface ²⁵⁶
DFG	943	1543	PPLN WG	86%–95%	Quantum interface ²⁵⁷
SFG	1550	863	PPTFLN	73%	Upconversion detection ²⁵⁸
DFG	780	1541	PPLN WG	95%	QFC ²⁵⁹

^aTD Ti, diffused waveguide; LN WG, thick film lithium niobate waveguide.^bIncluding all losses.

photons, confirmed by the observation of a Hong-Ou-Mandel (HOM)²⁶³ interference dip detection. The frequency conversion of the photons from two different quantum dots²⁴² was demonstrated using a single-frequency converter, enabling two-photon interference and highlighting the potential to mitigate fabrication-induced variations in quantum dot emission. Donohue et al.²⁶⁴ theoretically showed that engineering of the pump pulse leads to highly effective SFG conversion with the possibility for time-lensing and bandwidth compression. Sielberhorn's group presented the concept of a quantum pulse gate (QPG),²⁶⁵ using a spectrally engineered pump for SFG to select a particular temporal mode in an upconverted photon using PPLN. The QPG concept was further developed into temporal-mode tomography²⁶⁶ and temporal-mode demultiplexing.²⁶⁷ In addition, the third-order nonlinear effects are suitable for the task of single-photon shaping. Matsuda²⁶⁸ experimentally demonstrated a 0.4 THz single-photon spectral shift using cross-phase modulation (XPM) in dispersion-engineered photonic crystal fiber. All of these techniques can be implemented on the TFLN platform and, combined with electro-optic phase shaping techniques,^{269–271} form a versatile toolbox for spectral and temporal shaping.

4.1.3 Photon-pair generation

Lithium niobate may also be used as a source of photon pairs and single heralded photons via SPDC (Fig. 6). In the standard perturbative approach to SPDC within a single-mode waveguide and under low pump power, the generated quantum state can be approximated as

$$|\psi\rangle = |0,0\rangle_{s,i} + K \int d\omega_s d\omega_i f(\omega_s, \omega_i) \hat{a}_s^\dagger(\omega_s) \hat{a}_i^\dagger(\omega_i) |0,0\rangle_{s,i},$$

where heralding removes the vacuum component $|0,0\rangle_{s,i}$. Here, all relevant factors, including the nonlinear coefficient (e.g., the second-order susceptibility), have been absorbed into the constant K , and $\hat{a}^\dagger(\omega)$ is the photon creation operator at frequency ω . The joint spectral amplitude (JSA), $f(\omega_s, \omega_i)$, has a clear physical interpretation: the modulus squared of the normalized JSA, $|f(\omega_s, \omega_i)|^2$, defines the joint spectral probability density of the signal and idler photons. The JSA consists of two main components: the pump envelope function and the phase-matching function, the latter arising from momentum conservation. By engineering JSA, especially through the phase-matching conditions, it is possible to reduce spectral correlations, thereby increasing the purity of the state and enhancing interference visibility. For a comprehensive analysis, we refer the reader to a recent review on nonlinear engineering of desired photon characteristics.²⁷²

In contrast to the QFC application, photon-pair generation is widely associated with the use of TFLN devices together with Ti-diffused (TD) waveguides and even microresonators.^{86,273} Table 3 contains the comparison of different realized photon-pair sources in terms of pair generation rate (PGR), coincidence-to-accidental ratio (CAR), second-order autocorrelation at zero delay of a heralded single-photon $g_h^{(2)}(0)$, and two-photon interference experiment type and its visibility. The phase matching is limited not merely by QPM but also by phase matching among different spatial waveguide modes at

different mixing frequencies (modal phase matching, MPM). Introducing dual-layer TFLN (DLTFLN)²⁸³ and layer-poled TFLN (LPTFLN),²⁸⁴ it became possible to achieve nonzero TE_{00} to TE_{01} nonlinear mode overlap, improving the efficiency of MPM devices.

Table 3 shows that TFLN-based devices are already able to achieve from units^{86,243,279} up to several hundreds^{280,285} of GHz pair generation rates. In all papers where $g_h^{(2)}(0)$ was measured, its level goes below 0.2, which is at a comparable level demonstrated for silicon photonics.^{290,291} The purity and indistinguishability for all demonstrated devices are on a decent level ($>90\%$); however, only one paper²⁷⁸ demonstrated performance akin to the best silicon sources.^{290,291}

4.1.4 Squeezed state generation

One may also use the nonlinear processes to generate the squeezed states of light, where one of the quadratures achieves noise distribution lower than the shot noise level.²⁹² This property can help to improve the precision of gravitational wave detection,⁹ frequency metrology,²⁹³ resolution of imaging,²⁹⁴ or other applications,²⁹³ where the performance is limited by shot noise. The generation of a squeezed state requires phase-sensitive parametric amplification, which can be achieved using nonlinear optical effects.

The generation of squeezed states in lithium niobate waveguides was shown using proton exchanged,^{295–297} Ti-diffused,²⁹⁸ thick films,^{299,300} and microdisk³⁰¹ devices. Mondain et al. demonstrated the generation of the squeezed light in a single parametric down-conversion periodically poled waveguide made by SPE²⁹⁷ with a co-integrated interferometer. The generated light has -2.08 ± 0.05 dB squeezing and 2.80 ± 0.05 dB

anti-squeezing levels measured with respect to shot noise level. A total of 5 dB on-chip squeezing²⁹⁸ was generated in a resonant TD PPLN structure with active electrical tuning. The highest squeezing level achieved on a chip, to the best of our knowledge, is -9.7 ± 0.8 dB,²⁹⁹ shown in the thick film (5 μm) PPLN waveguide and, on the same platform, -6 dB squeezing over a 2.5 THz bandwidth.³⁰⁰

TFLN has also shown a capability for squeezed states with -0.56 ± 0.09 dB (inferred on-chip -2.6 dB) quadrature squeezing over 7 THz.³⁰² This was achieved in a Z-cut 600 nm periodically poled device. Using a silicon nitride strip-loaded waveguide on an X-cut TFLN,³⁰³ picosecond pulse squeezing was achieved with a squeezing of -0.33 ± 0.07 dB (-1.7 ± 0.4 dB on-chip inferred). The most broadband squeezer achieved in TFLN reaches 25 THz³⁰⁴ with -4.9 dB squeezing on-chip. In a recent paper, Arge et al.³⁰⁵ demonstrated an MPM TFLN resonator for squeezed state generation with -0.46 dB shot noise reduction.

4.2 Hybrid Multiplexing of Photon-Pair Sources

Developing a pure single-photon source is a “holy grail” of quantum photonics. A promising approach to achieve periodic, near-deterministic generation of pure single photons involves active multiplexing of photon-pair sources. The envisioned fully on-chip, hybrid spatial (M sources) and temporal (N modes) multiplexed photon source requires several key components (see Fig. 7):

- (i) high-quality photon-pair sources,
- (ii) filters to reject the pump and couplers to separate the signal and heralding photons,

Table 3 Summary of the demonstrated LN photon-pair sources.

Platform (Ref.)	PGR, $\frac{\text{GHz}}{\text{mW}}$	$g_h^{(2)}(0)$	CAR	Visibility (type)
SPE PPLN ²⁷⁴	0.0015	N/A	N/A	97% $\pm$ 1.84% (Franson)
LN WGMR ²⁷³	0.013	<0.2	N/A	N/A
Zn:PPLN ²⁷⁵	0.0015	0.1 ± 0.012	260,000 – 7.5	93% $\pm$ 17% (HOM)
PPTFLN ²⁷⁶	0.027 ^a	N/A	6900 ± 200	N/A
PPTFLN ²⁷⁷	0.008 ^a	N/A	631 ± 210	N/A
PPTFLN Ring ⁸⁶	2.6 ± 0.1^a	0.008 ± 0.0007	$14,682 \pm 4427$	N/A
PPTFLN ^{278b}	0.045	0.022 ± 0.004	$67,224 \pm 714$	99.3% $\pm$ 1.9% (Franson)
PPTFLN ²⁷⁹	13	N/A	100,000	98% (Michelson)
PPTFLN ²⁸⁰	279	N/A	599	99.17% (Franson)
PPTFLN ²⁸¹	0.024 ^a	N/A	N/A	96.3% (spectral purity)
PPLN ²⁸²	0.178	N/A	>8000	98.2% $\pm$ 0.3% (Franson)
PPTFLN ²⁴³	1.27 ^a	N/A	1182	N/A
MPM DLTFLN ²⁸³	41.77	<0.2	$58,298 \pm 1297$	N/A
MPM LPTFLN ²⁸⁴	0.34 ^a	0.008	663	N/A
PPTFLN ²⁸⁵	230 ± 50^a	N/A	N/A	100% $\pm$ 1% (Michelson)
PPLN TD ²⁸⁶	0.125 ^a	N/A	N/A	N/A
PPLN TD ²⁸⁷	11.86 ^a	N/A	N/A	N/A
MPM TFLN ²⁸⁸	0.672 ^a	N/A	N/A	N/A
MPM TFLN ²⁸⁹	1.44×10^{-6}	N/A	805 ± 5	N/A

^aCalculated from the paper data.

^bPresenting the best performance over multiple devices.

- (iii) high-efficiency SNSPDs,
- (iv) dedicated cryogenic electronics with low detect-to-switch latency,
- (v) very low-loss delay lines to compensate for this latency,
- (vi) fast optical switches with CMOS-compatible driving voltages, and
- (vii) very low-loss single photon buffers to relocate photons in a predetermined target time-bin.

This source is based on a logarithmic tree structure, where each photon traverses the (lossy) switch $\lceil \log_2(M) \rceil$ times, followed by temporal multiplexing.²⁴ TFLN offers two significant advantages. (i) Pump photons can be filtered out relatively easily due to their spectral separation from the generated photon pairs. (ii) TFLN modulators are capable of operating at cryogenic temperatures, albeit with a slight increase in modulation voltage compared with room temperature operation.³⁰⁶ On the other hand, cryogenic environments enable the use of superconducting electrodes, which further enhance the performance of LN-based modulators.³⁰⁷

On-chip photon-pair sources can be quasi-phase-matched type-II PPLN waveguides, offering high spectral purity.²⁸¹ These sources enable the use of high-efficiency polarization beam splitters to separate heralding and heralded photons directly on-chip.²⁰⁴ However, achieving sufficient pump rejection, typically on the order of 100 dB, poses a significant challenge. To address this, Bragg filters adapted to TFLN³⁰⁸ can be used, and the large spectral separation between the pump and photon pairs improves filtering performance. Once filtered, heralding photons can be detected using

waveguide-integrated SNSPDs (see Sec. 3.2), at cryogenic temperatures, activating a dedicated heralding logic. A critical step is the implementation of low-latency logic, requiring heralded photons to be delayed for switch activation within the same clock cycle. Most implementations of multiplexing schemes suffer from electronic latencies of around 100 ns.²⁰⁰ Notably, a very recent study demonstrated a cryogenic operation of a dedicated logic, achieving latency as low as 23 ns,³⁰⁹ corresponding to a 4.18 dB loss for TFLN-based delay lines (~ 0.182 dB/ns for state-of-the-art propagation losses of 1.3 dB/m⁹⁵). Latency-induced loss sets an upper bound on the achievable source efficiency at roughly 0.38, notably below the 2/3 efficiency threshold suggested for loss-tolerant linear optical quantum computing scheme based on the stabilizer tree graph codes.³¹⁰ Therefore, minimizing latency is critical as both loss and decoherence scale with the length of the delay lines. Achieving latencies on the order of just a few nanoseconds would be necessary to limit losses in TFLN-based delay lines to acceptable levels. Encouragingly, tailored cryogenic control circuits are currently under active development (see Ref. 311 for a comprehensive review). Such advancements enable a new generation of on-chip, scalable, high-quality single-photon sources that surpass the efficiency threshold necessary for practical implementations of linear optical quantum computing. In addition, to perform single-photon multiplexing to the desired output mode, CMOS-compatible modulators are required, and they have already been reported.⁶² Finally, regarding footprint, low-loss components have been fabricated at the wafer scale on 6-in. TFLN wafers.¹¹¹ Therefore, it is,

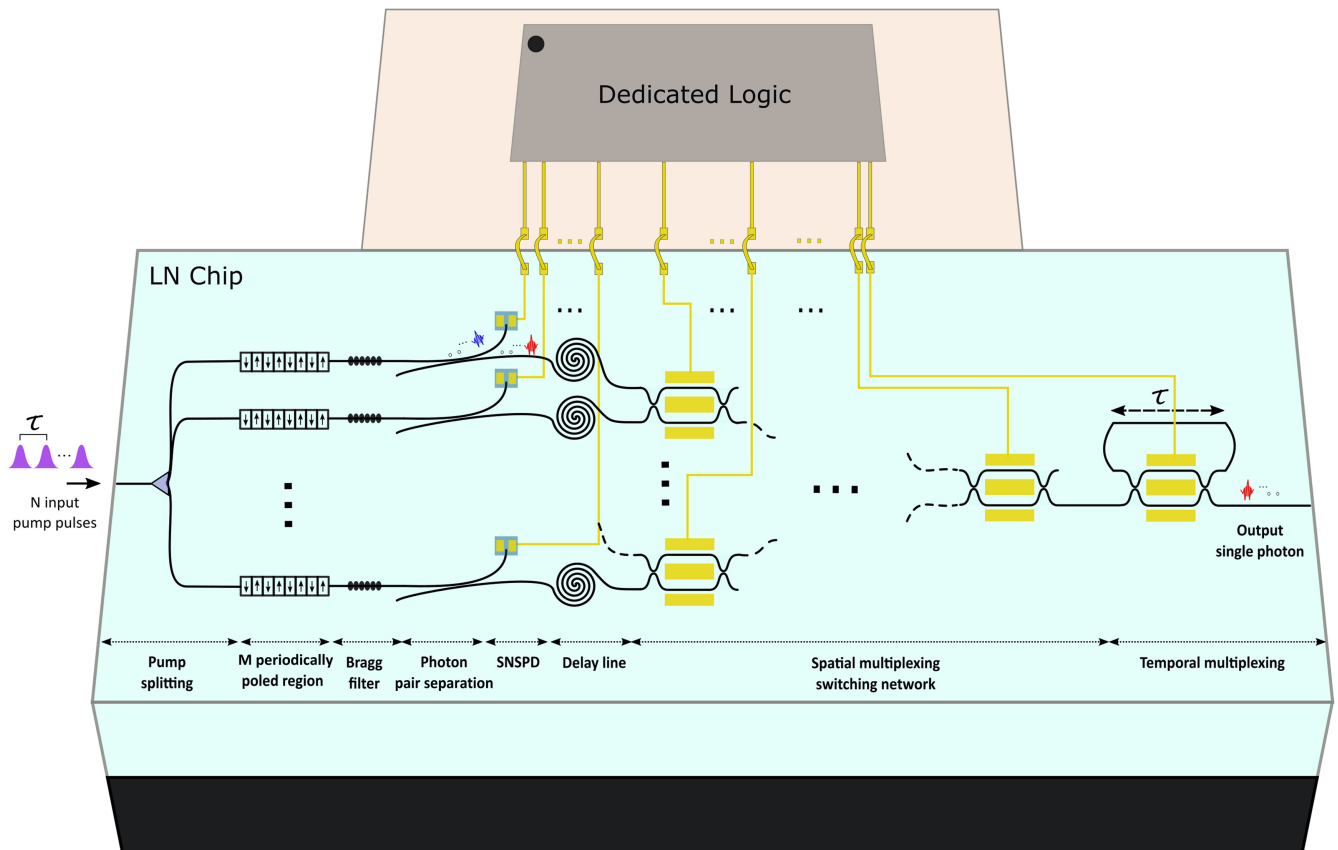


Fig. 7 Envisioned hybrid multiplexing scheme on the TFLN chip with a dedicated cryogenics logic.

in principle, possible to achieve efficient on-chip multiplexing integrated with all necessary components, which can also be volume-fabricated.

The proposed TFLN-based hybrid multiplexed source, using the current state-of-the-art parameters, does not yet reach the efficiency of the leading QD source, which has achieved efficiencies as high as 0.712 at 885 nm.³¹² However, QD sources face scalability challenges due to poor visibility when interfering with single photons generated by independent quantum dots. Although this issue can be addressed using photon shaping techniques, as discussed in Sec. 4.1.2, scaling QDs remains difficult in practice. As most quantum protocols require multiple indistinguishable single-photon inputs across different modes simultaneously, generating such a stream typically mainly relies on a single QD combined with a demultiplexing strategy,³¹³ which reduces operation rate and introduces additional losses. On the other hand, high-rate nonlinear single-photon sources feature excellent indistinguishability between sources by coherent pumping,³¹⁴ leading to scalable quantum systems. Furthermore, the emission wavelength of single photons from pair sources can be flexibly engineered through dispersion engineering, enabling generation directly in the telecommunication band, crucial for integration with existing fiber infrastructure.

4.3 Quantum Networks

The generation of remote entanglement between spatially separated quantum nodes is foundational to quantum networks, enabling applications ranging from distributed quantum sensing³¹⁵ to distributed quantum computing.³¹⁶ Quantum memories are crucial components in these networks as they can become entangled across distant nodes.²⁵⁵ However, current entanglement rates between quantum memory nodes are below 1 Hz, which limits scalability.³¹⁷ One approach to increase these rates is to multiplex memory nodes, which requires scalable, large-scale switch networks with low insertion losses to support multiple memory channels while maintaining high efficiency. In addition to efficient multiplexing, interfacing these memory nodes with an optical fiber might demand the ability to shift photon frequencies with high efficiency. Precise frequency manipulation can be achieved via: (i) QFC, discussed in Sec. 4.1.2, or (ii) electro-optic spectral shearing (EOSS).^{269–271} EOSS shifts

the photon frequency by applying a linearly ramped phase modulation, $\Delta\phi = -Kt$, which translates a photon's initial frequency ω to a new frequency $\omega + K$, allowing frequency shifts on the order of ~ 100 GHz. Recent advances in TFLN platforms provide a promising solution for multiplexing quantum nodes. These platforms integrate low-insertion-loss couplers (<1 dB per facet), frequency shifters (capable of ~ 33 GHz shifts), and a 2×2 switching network (with extinction ratios exceeding 20 dB) on a single chip, operating from visible to near-infrared wavelengths³¹⁸ [see Fig. 8(a)]. These advancements are expected to enhance entanglement rates and pave the way for burgeoning quantum networks.

On the other hand, an all-photon quantum repeater eliminates the need for matter-based quantum memories but requires large entangled resource states (i.e., graph states),¹⁵ which, in principle, can be generated deterministically via quantum emitters³²⁰ or by multiplexing probabilistic linear optical entangling gates. To achieve determinism, a feedforward mechanism is necessary, which in turn requires ultra-low loss and low-latency control.

4.4 Quantum Metrology

High-fidelity state generation and manipulation may also be used for quantum sensing applications. For quantities such as phase or temperature, which lack direct self-adjoint operators, values can be estimated by measuring related observables, known as parameter estimation. This process is essential in applications such as gravitational wave detection,⁹ where phase shifts produced by gravitational waves can be estimated through interference. Squeezed states of light are increasingly promising in that sense as they can be bright and enhance sensitivity by reducing the noise level in their squeezed quadrature (see Sec. 4.1.4).^{17,293} Recent advancements have demonstrated the integration of squeezed light sources into photonics circuits on the same TFLN chip, achieving nearly complete phase sensor functionality with quantum-enhanced sensitivity³¹⁹ [see Fig. 8(b)].

4.5 From NISQ to the Next Frontier in Quantum Computing

It is strongly believed that raw quantum computational power, which surpasses the performance of classical supercomputers on

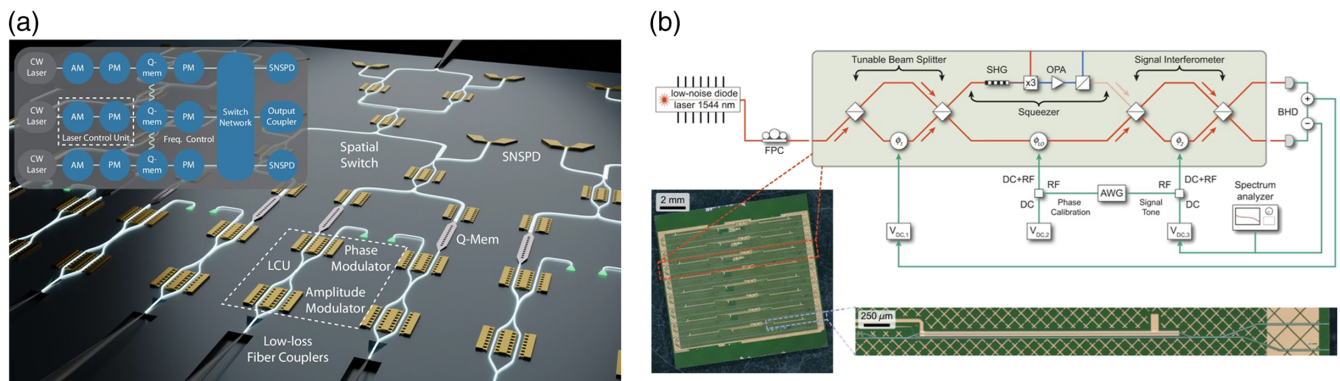


Fig. 8 TFLN-based quantum node and a quantum sensor: (a) A proposed quantum node including matter-based quantum memory. Figure adapted with permission from Ref. 318 © 2024 The Author(s). (b) Quantum-enhanced phase sensor utilizing squeezed light. Adapted with permission from Ref. 319 © 2023 The Author(s).

specific tasks, can be demonstrated with modest experimental requirements via rudimentary quantum computers. In computational complexity terms, certain abstract problems that are intractable, if not impossible, for classical computers can be solved by these quantum devices. Photonics offers a candidate for possible demonstration of the so-called quantum computational advantage: boson sampling (BS).³²¹ It is arguably one of the simplest NISQ models in terms of technological requirements as it involves preparing a quantum state with indistinguishable photons, letting it evolve in a multimode optical network through quantum interference, and then detecting the output; see Fig. 9(a) for a conceptual scheme. The distribution of the detection pattern approximates the probability distribution governing the experiment, which is difficult for classical computers to sample. This is because calculating the probability corresponding to an output event requires estimating the permanence of a complex-valued matrix, a task considered notoriously difficult to compute for large matrices. Note that surpassing classical computers depends significantly on the number of occupied input modes (n) and the total number of interferometer modes (m);³²³ however, increasing either of these is challenging in practice.³²⁴ Over the last few years, some plausible demonstrations of quantum advantage with Gaussian states of light³²⁵ have been reported.^{1,314,326} However, debates continue regarding these claimed demonstrations as photon losses and partial distinguishability may lead to classically simulable quantum devices.^{327,328} In addition to claimed demonstrations of quantum advantage, BS can also be mapped to the simulation of molecular dynamics^{329,330} and graph theoretical calculations²³ and can be used to enhance classical algorithms.³³¹ A more comprehensive review of BS can be found in Ref. 332.

TFLN photonics is highly effective in generation of quantum states of light and provides precise photon path manipulation due to its high electro-optic coefficients and low-loss properties.

These capabilities are expected to pave the way for new demonstrations of quantum advantage. A two-photon quantum interference that occurs at a 2×2 beam splitter represents the most basic scenario and can be further generalized to n -photon quantum interference in an m -mode unitary, called multimode quantum interference. It is crucial for quantum technologies and can be realized using a series of interconnected MZIs. An MZI equipped with a phase shifter in one of its arms acts as a tunable beam splitter, allowing for precise control over photon pathways. This functionality is essential for constructing quantum circuits with complete reconfiguration capabilities. Once the unitary evolution U for a specific quantum protocol is defined, it can be translated into a physical circuit by decomposing U into layers of tunable MZIs and phase shifters. In photonics, two widely used universal decomposition methods are the Reck (triangle) scheme³³³ and the Clements (rectangular) scheme.³³⁴ Recent studies have demonstrated a TFLN-based integrated optical circuit that serves as a fast and reconfigurable four-mode universal linear quantum processor, consisting of a series of MZIs and phase shifters that are programmable through the electro-optic effect. One study followed the Clements scheme and interfaced it with deterministic solid-state single-photon sources based on quantum dots in nanophotonic waveguides at 930 nm,¹⁹⁸ as shown in Fig. 9(b), whereas another study implemented the Reck scheme,³²² as shown in Fig. 9(c), highlighting the flexibility and scalability of on-chip quantum state manipulation using TFLN technology.

Rudimentary quantum computers help build a foundation of techniques and technologies that will eventually support progress toward the “holy grail” of quantum information science: a universal fault-tolerant quantum computer. Universality refers to a computer capable of performing arbitrary conceivable processes allowed by quantum mechanics, whereas fault tolerance supplements error correction codes with protocols for

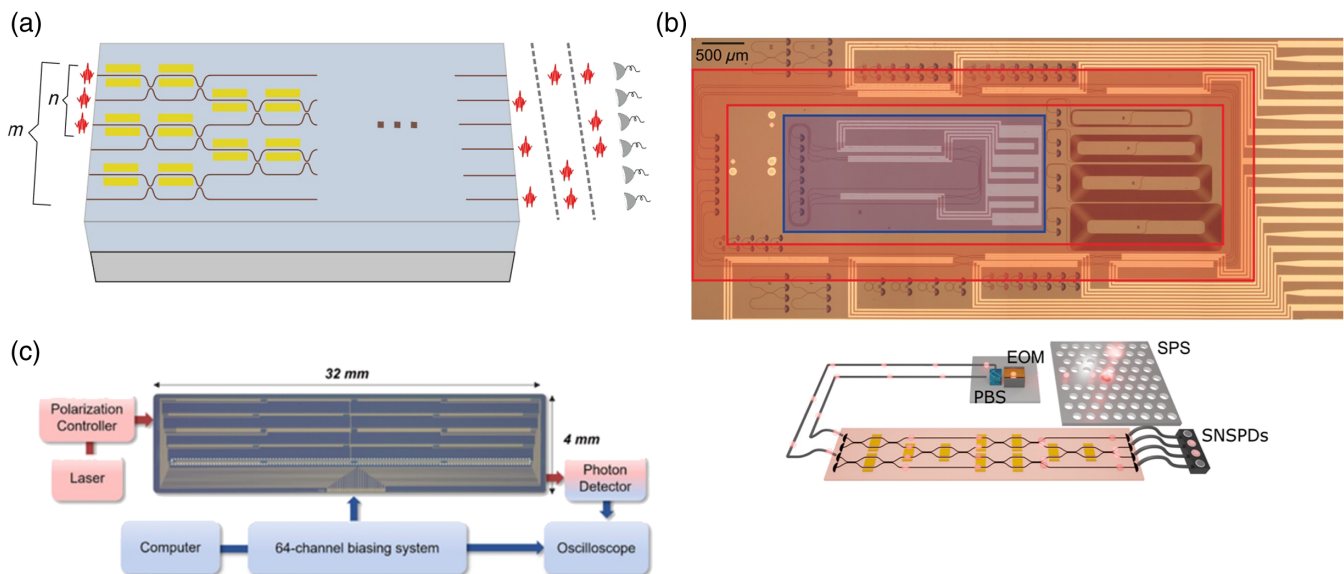


Fig. 9 Universal linear optics based on TFLN. (a) Conceptual scheme of a BS: m -mode reconfigurable linear interferometer is fed with n single photons, after a number of trials output statistics are built through the coincidence detection. (b) A four-mode interferometer following Clements decomposition. Adapted with permission from Ref. 198 © 2023 American Association for the Advancement of Science (AAAS). (c) A four-mode interferometer following Reck decomposition. Adapted with permission from Ref. 322 © 2023 The Author(s).

reliable computation. One of the primary approaches to realizing such a computer relies on discrete variables, i.e., qubits. Qubits are two-level quantum systems, i.e., $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ where $\alpha, \beta \in \mathbb{C}$, called probability amplitudes satisfying normalization condition $|\alpha|^2 + |\beta|^2 = 1$. The basis $|0\rangle$ and $|1\rangle$ represent the mutually distinguishable states of the system. Photonic architectures for encoding qubits can be based on various degrees of freedom of the single photon, such as polarization, transverse spatial modes, frequency modes, path, and time-bin degrees of freedom. Among these, path encoding, where a photon occupies different optical paths (waveguides), is central to integrated photonics because it allows for extremely low-error qubit operations using Mach–Zehnder interferometers and phase shifters, as noted above.

TFLN photonics is an emerging candidate for universal optical quantum computation. The standard gate-based model,³³⁵ which relies on realizing deterministic gates through ancillary circuitry and photons, faces polynomial yet prohibitive overheads at a practical level. A new paradigm, so-called measurement-based quantum computation (MBQC), pioneered by Raussendorf and Briegel³³⁶ changed the way we look at quantum computation by offering a significantly more resource-efficient approach, shifting the focus from the gate-based model to the generation of highly entangled resource states, e.g., universal 2D cluster states.^{337,338} In MBQC, computation is driven by performing single-qubit measurements, which are straightforward to implement on photonic qubits, on these entangled states. Depending on the complex correlation shared between qubits, local measurements on one qubit influence its neighbors. The inherent randomness of these outcomes is compensated through adaptive measurements on neighboring qubits via active feed-forwarding, enabling the effective execution of a quantum algorithm through precise coordination. A linear optical implementation of MBQC, primarily developed by Browne and Rudolph, significantly reduced the resources required to generate large photonic cluster states³³⁹ by introducing the fusion mechanism, two-qubit measurements between qubits from separate entangled states. Subsequent theoretical advancements, incorporating ideas from percolation theory, demonstrated that universal (incomplete) lattice resource states could be produced from 3-qubit GHZ states³⁴⁰ using fusion operations.^{341–343} More recently, the fusion-based quantum computing scheme²⁰¹ has emerged, utilizing many copies of small entangled resource states and fusion operations. However, generating resource states, including 3-qubit GHZ states, with linear optics remains challenging, as all interferometers for two-photon gates are probabilistic. To address this limitation, entangling gates can be multiplexed to deterministically generate resource states.³⁴⁴ High-speed, low-loss optical switches enabled by TFLN technology are expected to herald a new era in the deterministic generation of resource states using linear optics. On the other hand, based primarily on Lindner and Rudolph’s proposal,³⁴⁵ quantum emitters have been shown to be promising candidates for the deterministic generation of resource states, including examples such as QDs³⁴⁶ and single atoms.³⁴⁷ However, the most concerning issue—apart from the increased complexity—is that these methods are vulnerable to stochastic noise as opposed to the linear optical-based methods.

In optical CV MBQC architectures,³⁴⁸ squeezed light is the main ingredient. Ideally, cluster states can be generated by entangling neighboring infinitely squeezed states, which are, however, nonphysical. In addition, homodyne detections,

depending on the local oscillator’s phase, drive the computation. In practice, approximate CV cluster states—prepared using finite squeezing rather than ideal infinite squeezing—can be generated deterministically, and homodyne measurements can be effectively implemented. The use of approximate cluster states introduces computational noise resulting from finite squeezing, which can lead to errors in computation. Fortunately, as in other quantum computing platforms, such errors can be mitigated through quantum error correction.

To address scalability challenges, the frequency multiplexing³⁴⁹ and the temporal multiplexing³⁵⁰ schemes have been suggested. Frequency multiplexing schemes encode quantum information in the quadratures of optical frequency combs,³⁵¹ whereas temporal multiplexing schemes encode information in the quadratures of fields at different time bins.³⁵² In Ref. 352, a 2D cluster state with a cylindrical topology was successfully generated using an experimental setup comprising a pair of bow-tie-shaped squeezers, beam splitters, and optical fiber-based delay lines. Although this scheme enables scaling computation with a fixed number of spatial resources, it also has inherent limitations that constrain the achievable computation depth. Specifically, in temporal encoding schemes, the maximum allowable delay length is limited by propagation losses, which introduce additional computational noise. Beyond a certain delay length, depending on the quantum error correction scheme employed, fault-tolerant computation becomes infeasible.

In Ref. 351, the quantum-wire cluster state in the quantum optical frequency comb is created by pumping optical parametric oscillators with synchronous phase-locked polychromatic beams. A natural next step is to extend this concept to chip-scale implementations using low-loss microresonator platforms such as SiN. As a foundational requirement, squeezed state generation on the SiN platform has been demonstrated.^{353,354} Following this, recently, SiN has been used to generate rich-structured CV cluster states across quantum frequency combs using multiple pump lines.^{355,356} However, the squeezing levels currently demonstrated on SiN are still below the 12 to 15 dB range considered necessary for fault-tolerant CV quantum computing^{357–359} due to the platform’s reliance on inherently weak third-order nonlinearities. Moreover, frequency multiplexed cluster states require polychromatic homodyne detection,³⁶⁰ where each frequency mode is individually addressed. This increases the experimental complexity and overhead, making the approach more challenging in practice in comparison to time-domain approaches.

On the other hand, TFLN photonics offers a compelling alternative to bring approximate CV cluster states closer to their ideal counterparts by combining low propagation losses comparable to SiN with strong second-order nonlinearities (see Sec. 4.1.4). Its support for increased bandwidth may facilitate higher repetition rate setups, which, in turn, would allow more temporal modes to fit within the maximum allowable delay length, ensuring effective quantum error correction. Furthermore, a TFLN-based temporally multiplexed compact CV cluster state generator could further enhance system performance by minimizing coupling losses. Loop-based architectures with integrated switches, as proposed in Ref. 361, could further benefit from TFLN photonics, making it a key technology for the realization of CV MBQC. Furthermore, as CV MBQC is driven by phase-dependent homodyne detection, fully on-chip

implementation requires programmable local oscillators where TFLN stands out.³⁶² Taken together, these capabilities position TFLN as a strong candidate for scalable, fault-tolerant CV MBQC.

To further clarify the motivation for using TFLN over SiN, we include Table 4, which summarizes the key differences between the two platforms in the context of large-scale CV MBQC. Although Table 5 provides a material-level comparison, Table 4 highlights the application-specific differences relevant to scalable CV MBQC. Though each material platform alone has its own advantages and disadvantages, lithium niobate is particularly highlighted featuring both high second- and third-order nonlinearity, ultra-low propagation loss, high EO coefficient with pure phase modulation, and high level of squeezing, which are keys for quantum photonic applications. It is also obvious that to achieve a TFLN chip-level quantum photonic system, the heterogeneous integration of QD sources and detectors is a necessary step, as discussed in Sec. 3. Furthermore, the heterogeneous integration of TFLN with silicon nitride platform would lead to a further technological breakthrough leveraging both extremely low loss of silicon nitride and all distinct features of lithium niobate.

5 Challenges and Outlook

Over the past few years, TFLN quantum photonics has emerged as a versatile integrated platform for developing future quantum technologies. Its low-loss, ultra-fast electro-optic switching capabilities, and high nonlinear coefficients are key characteristics that make it ideal for quantum photonic applications. These applications include enhanced quantum metrology, near-deterministic single-photon sources through active feedforwarding, high-secret-key-rate QKD, high-rate boson sampler, and efficient resource generation for universal computation and the quantum internet. The ultimate goal would be to realize fault-tolerant fully on-chip universal quantum processors to minimize coupling losses between different platforms, thereby significantly enhancing efficiency and scalability. However, the integration of quantum light sources, large-scale circuits, and photon detectors on a single chip remains a challenge. This overview lists the current challenges in TFLN quantum photonics and explores potential pathways toward achieving fully on-chip quantum information processing.

- Sub-dB/m propagation losses: Propagation losses in TFLN have seen remarkable progress, recently achieving values as low

Table 4 Comparison between SiN and TFLN platforms for large-scale CV MBQC.

Feature	SiN	TFLN
Cluster encoding	Frequency domain	Well-suited for the temporal domain
Pump configuration	Polychromatic	Monochromatic
Detection scheme	Polychromatic homodyne	Standard homodyne
Scalability	Limited by squeezing level and mode complexity	Favorable via temporal multiplexing

Table 5 Comparison of different photonic material platforms.^a

Material	n_o (n_e)	α_{prop} , dB/m	Second-order nonlinear coefficient, pm/V	n_2 , m ² /W	EO coefficient, pm/V	Sources ^e	Detectors ^e	Squeezing max. ^f , dB
LiNbO ₃	2.21 (2.14)	1.3 ⁹⁵ 0.34 ⁹⁷	d_{33} : -25.2 (1064 nm) ³⁷	1.8×10^{-19} (1550 nm) ²⁰⁶	r_{33} : 30.9 (633 nm) ²²⁹	N/A	N/A	-11 ³⁰⁴
LiTaO ₃	2.119 (2.123)	5.6 ³⁶³	d_{33} : 13.8 (1064 nm) ³⁷	1.7×10^{-19} (800 nm) ^{e229}	r_{33} : 30.5 (633 nm) ²²⁹	N/A	N/A	—
Si	3.48	6.5 ³⁶⁴	N/A	5×10^{-18} (1550 nm) ²⁰⁶	N/A	N/A	✓	-4.17 ⁹²⁹⁰
Si ₃ N ₄	2	0.39 ²⁹¹	N/A	2.5×10^{-19} (1550 nm) ²⁰⁶	N/A	N/A	N/A	-8 ^{b354}
4H-SiC	2.56 (2.61)	9 ³⁶⁵	d_{33} : -11.7 (1064 nm) ³⁶⁶	7×10^{-19} (1550 nm) ^{c367}	r_{13} : 0.3 to 0.7 (1550 nm) ³⁶⁸	✓	✓	—
AlGaAs ^d	3.3	20 ³⁶⁹	d_{41} : 180 (1550 nm) ³⁶⁹	26×10^{-19} (1550 nm) ³⁷⁰	r_{41} : -1.5 (1520 nm) ³⁷¹	✓	✓	-21 ³⁷²

^aAll refractive index values are given at 1550 nm. For the second-order coefficient, only the highest tensor component in contracted notation d_{ij} is listed.

^bSqueezed states are generated in a microresonator structure.

^cThe lowest value between ordinary and extraordinary axes.

^dParameters of Al_xGa_{1-x}As are composition dependent.

^eThe presence of deterministic single-photon sources and single-photon detectors without heterogeneous integration.

^fThe lowest achieved squeezing level demonstrated on-chip in traveling wave configuration, unless stated otherwise.

^gRecalculated from the maximum measured squeezing value $|\tanh \zeta|^2 = 0.2$.

as 1.3 dB/m. Despite this achievement, there remains significant potential for further improvement as the theoretical material limit is ~ 0.1 dB/m.^{142,373} Achieving such ultra-low losses would result in the realization of ultra-low-loss delay lines, and by extension, ultra-low-loss recirculating loops (or binary division networks). This could, together with ultra-low-loss components, pave the way for the next generation of hybrid-multiplexed near-deterministic single-photon sources. To bridge this gap, ongoing efforts focus on refining fabrication techniques to reduce surface and sidewall roughness and developing advanced etching methods to further minimize losses.

- **Pump rejection:** On-chip pump rejection is crucial for preventing the generation of spurious photons within the network and, more importantly, for enabling the integration of SNSPDs on the same chip. Bragg filters in TFLN waveguides have demonstrated a 25 dB extinction using a 250 μm long periodic Bragg waveguide,³⁰⁸ though this performance was limited by high propagation losses (76 dB/cm). To achieve the desired extinction of over 100 dB, longer Bragg filters with significantly lower propagation losses are necessary. Innovations from other photonic platforms such as cascaded structures,^{374,375} hybrid approaches combining Bragg reflectors and ring resonators,^{376,377} and novel filter designs³⁷⁸ could provide promising avenues for improving pump rejection performance in TFLN.

- **Laser integration:** Integration of on-chip electrically pumped, power-efficient, and stable light sources is a challenge; however, it provides access to weak coherent state-related applications, such as future QKD links. This can be achieved through micro-transfer printing of III-V devices on the TFLN platform.

- **Photon-number resolving (PNR) detection:** PNR detection is critical for a variety of quantum photonic applications, such as heralding single-photon states, and by extension eliminating multiphoton contributions in photon-pair source multiplexing setups, and facilitating advanced protocols like Gaussian boson sampling.³²⁵ Transition-edge sensors implemented on LN waveguides have demonstrated PNR capabilities,³⁷⁹ but their response times, on the order of a few microseconds, limit their usage. To address this limitation, the concept of multiplexing multiple threshold SNSPDs can be adopted.³⁸⁰

- **Fiber-to-chip coupling:** Fiber-to-chip coupling is one of the limiting factors for TFLN applications, e.g., nonlinear conversion efficiency, upconversion detectors or other applications where coupling losses lead to a significant decrease in efficiency. However, the recent approach of trident edge couplers³⁸¹ demonstrated its potential to achieve a complex broadband transmission spectrum and 1.4 dB losses, which can be further optimized to a sub-decibel level. Alternatively, significant progress has been achieved in double-etched tapered edge couplers,¹⁷³ demonstrating low polarization sensitive sub-decibel coupling loss to an ultra-high NA fiber.

There are ongoing studies addressing all of these challenges, and, as discussed in the sections above, significant progress has been made in recent years. Despite these improvements and identified pathways, the realization of fully on-chip, near- and long-term quantum processors demands continued efforts in device fabrication, engineering, and monolithic integration. Nevertheless, based on our analysis and expectations, TFLN is poised to shape the future of integrated quantum photonics with its unique combination of performance, versatility, and scalability.

Code and Data Availability

This review does not involve the generation of new data or code. All data discussed were obtained from previously published sources, which are appropriately cited within the paper.

Acknowledgments

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